United States Patent Application Publication	20250276750
Kind Code	A1
Publication Date	September 04, 2025
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BICYCLE LOCK CARRIER ASSEMBLY

Abstract

A bicycle lock carrier assembly includes a bicycle lock carrier including a resting surface that includes a lock receiving member and one or more mounting provisions for securing to a frame of a bicycle, and a bicycle lock including a lock that fits inside the lock receiving member. The lock includes a lock member that is arranged to engage a locking structure of a portion of the bicycle lock which is outside the lock receiving member when the bicycle lock is placed on the resting surface, so as to lock the bicycle lock to the lock receiving member of the bicycle lock carrier.

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Appl. No.: **18/591197**

Filed: **February 29, 2024**

Publication Classification

Int. Cl.: **B62H5/00** (20060101)

U.S. Cl.:

CPC **B62H5/00** (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] This invention relates generally to bicycle locks, and particularly to a carrier for a bicycle lock, such as a folding lock, which is lockable to the carrier.

BACKGROUND OF THE INVENTION

[0002] Folding locks, for locking bicycles to a post or other object, are made up of a series of metal bars (flat plate bar links), which are linked together by joints, typically rivets. The rivets allow the bar links to rotate so they can be folded into a compact configuration for storage on the bicycle and folded out for fastening the bicycle to the object.

[0003] Folding locks are normally stored on a carrier, which is typically secured to a portion of the bicycle frame. It is not known in the prior art to lock the folding lock to the carrier itself.

SUMMARY

[0004] The present invention seeks to provide an improved carrier for a bicycle lock, particularly, a folding lock. The carrier of the invention may be used for bicycles and other items, such as but not limited to, strollers, carriages and the like, as is described in detail further hereinbelow. The terms “folding lock” or “bicycle lock” or “folding bicycle lock” are used to encompass a lock for bicycles and these other items. Unlike the prior art, the bicycle lock carrier of the invention enables the folding lock to be locked to the carrier, which among other things, prevents theft of the folding lock from the bicycle.

[0005] There is thus provided in accordance with a non-limiting embodiment of the invention a bicycle lock carrier assembly including a bicycle lock carrier including a resting surface that includes a lock receiving member and one or more mounting provisions for securing to a frame of a bicycle, and a bicycle lock including a lock that fits inside the lock receiving member, the lock including a lock member that is arranged to engage a locking structure of a portion of the bicycle lock which is outside the lock receiving member when the bicycle lock is placed on the resting surface, so as to lock the bicycle lock to the lock receiving member of the bicycle lock carrier.

[0006] In accordance with a non-limiting embodiment of the invention, the bicycle lock includes a folding bicycle lock that includes a plurality of link members which are pivotally connected to one another by joint members, and the locking structure of the portion of the bicycle lock is located on a lockable link member, which is one of the link members.

[0007] There is thus provided in accordance with a non-limiting embodiment of the invention a bicycle lock carrier assembly including a bicycle lock carrier including a resting surface that includes a lock receiving member and one or more mounting provisions for securing to a frame of a bicycle, and a bicycle lock including a lock that fits inside the lock receiving member, wherein the bicycle lock has a configuration in which the bicycle lock is locked to the bicycle lock carrier and blocks access to the one or more mounting provisions.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] The present invention will be understood and appreciated more fully from the following detailed description, taken in conjunction with the drawings in which:

[0009] FIG. **1** is a simplified perspective illustration of a bicycle lock carrier, in accordance with a non-limiting embodiment of the present invention.

[0010] FIGS. **2**, **3**, **4**, **5**, and **6** are simplified rear view, top view, bottom view, left side view, and right side view illustrations, respectively, of the bicycle lock carrier.

[0011] FIG. **7** is a simplified perspective illustration of a folding bicycle lock about to be inserted into the bicycle lock carrier of the invention.

[0012] FIG. **8** is a simplified perspective illustration of the folding bicycle lock inserted into the bicycle lock carrier of the invention, and prior to be locked therein.

[0013] FIG. **9** is a simplified perspective illustration of the folding bicycle lock locked to the bicycle lock carrier of the invention.

[0014] FIGS. **10** and **11** are simplified perspective illustrations of the bicycle lock carrier of the invention secured to a bicycle, respectively before and after locking the folding lock to the carrier.

DETAILED DESCRIPTION

[0015] Reference is now made particularly to FIG. **1**, as well as to FIGS. **2-6**, which illustrate a bicycle lock carrier **10**, in accordance with a non-limiting embodiment of the present invention.

[0016] The bicycle lock carrier **10** includes a resting surface **12**, a first end **14** of which may be curved upward (in the sense of the drawings) and a second end **16** of which may include a lock receiving member **18**. In the non-limiting illustrated embodiment, the resting surface **12** may be rectangular, but other shapes are within the scope of the invention. Resting surface **12** may be smooth or have any other kind of texture, such as being formed with parallel ridges, knurling and other textures. Resting surface **12** may be formed with one or more mounting provisions **20**, such as holes for passing therethrough fasteners (such as screws) for securing the bicycle lock carrier **10** to a portion of a bicycle frame.

[0017] The first end **14** may be curved to correspond to the curvature of link members of a folding bicycle lock, as will be described below.

[0018] The lock receiving member **18** may be a cylindrical member with an inner, hollow, circular chamber **17** in which the user can place the lock of the folding bicycle lock, as will be described below. The lock receiving member **18** may include an outer curved abutment **19** that couples to resting surface **12**. The outer curved abutment **19** may be curved to correspond to the curvature of the link members of the folding bicycle lock, as will be described below. The lock receiving member **18** may have a recess **13** formed at an end **15** thereof. The recess **13** may be shaped to allow passage of a locking structure (of one of the link members of the folding bicycle lock) into inner chamber **17**, as described below.

[0019] Resting surface **12** may be formed with one or more lateral lock stop members **22**, against which the folding bicycle lock may abut when placed in bicycle lock carrier **10**. Resting surface **12** may be formed with one or more bicycle frame stop members **24**, against which the frame of the bicycle may abut when the bicycle lock carrier **10** is secured to the bicycle.

[0020] The bicycle lock carrier **10** may be made of a steel alloy or any other sturdy material.

[0021] Reference is now made to FIG. **7**, which illustrates a folding bicycle lock **30** about to be inserted into the bicycle lock carrier **10**. The folding bicycle lock **30** may include a plurality of link members **32**, which are pivotally connected to one another by joint members **34**, such as rivets or pins. Link members **32** may be made of a material hardened and dimensioned against cutting or other vandalistic forces (such as a hardened steel alloy or any other suitable material). Link members **32** may be covered with a non-metallic covering, such as a plastic, rubber, silicone or other polymer coating or covering, which is aesthetically pleasing and prevents scraping or scratching the frame of the bicycle.

[0022] One of the link members **32** may be connected to a locking device, such as a lock **36**, which may be a cylinder lock operated by a key **38** or any other type of lock, such as but not limited to, a combination lock, cylinder lock, wafer lock, or wireless communication lock (that operates with a transponder that communicates with identification circuitry in the lock to gain authorized access to the lock), e.g., an RFID lock, NFC lock, Bluetooth lock, Wi-Fi lock, mobile device, and others. Other locking types can also be used. Another link member **32** is connected to a lockable link member **40**, which is typically longer than the rest of the link members **32**.

[0023] The lock **36** includes a lock member **42** that can engage a locking structure **43** (shown in FIG. **9**) of lockable link member **40** so as to lock the lock member **42** to the lock **36**. Without limitation, locking structure **43** may be a lug or notch or other structure. Opening the lock **36** releases the lock member **42** to unlock lockable link member **40** from lock **36**, so as to permit the user to unfold and stretch out the link members **32**.

[0024] Reference is now made to FIG. **8**, which illustrates folding bicycle lock **30** inserted into the bicycle lock carrier **10**, and prior to locking therein. It is seen that the ends of the link members **32** near the lock **36** fit against outer curved abutment **19** and the opposite ends fit against the inner surface of first end **14**. The lock **36** sits in inner chamber **17** up to, or close to, end **15** of lock receiving member **18**.

[0025] In FIG. **9**, the end of lockable link member **40**, which has the locking structure **43**, has been pivoted to align with lock **36** so that lock **36** can lock the locking structure **43**, thereby locking the folding bicycle lock **30** to the bicycle lock carrier **10**. The lockable link member **40** is positioned against or near the outside of end **15** of lock receiving member **18** and recess **13** allows the locking structure **43** to pass into inner chamber **17**. Since lockable link member **40** is on the outside of end **15** of lock receiving member **18** and lock **36** is on the inside of end **15** of lock receiving member **18**, folding bicycle lock **30** is now locked to lock receiving member **18** of bicycle lock carrier **10**. Only an authorized user with key **38** can now remove folding bicycle lock **30** from bicycle lock carrier **10**.

[0026] Reference is now made to FIGS. **10** and **11**, which illustrate the bicycle lock carrier **10** secured to a bicycle **50**, respectively before and after locking the folding lock **30** to the carrier **10**. It is noted that any screws used to secure bicycle lock carrier **10** to the frame of bicycle **50** are hidden from view and access by the folding lock **30**.

Claims

1. A bicycle lock carrier assembly comprising: a bicycle lock carrier comprising a resting surface that comprises a lock receiving member and one or more mounting provisions for securing to a frame of a bicycle; and a bicycle lock comprising a lock that fits inside said lock receiving member, said lock comprising a lock member that is arranged to engage a locking structure of a portion of said bicycle lock which is outside said lock receiving member when said bicycle lock is placed on said resting surface, so as to lock said bicycle lock to said lock receiving member of said bicycle lock carrier.

2. The bicycle lock carrier assembly according to claim 1, wherein said bicycle lock comprises a folding bicycle lock that comprises a plurality of link members which are pivotally connected to one another by joint members, and said locking structure of the portion of said bicycle lock is located on a lockable link member, which is one of said link members.

3. The bicycle lock carrier assembly according to claim 2, wherein an end of said resting surface is curved to correspond to a curvature of said link members of said folding bicycle lock.

4. The bicycle lock carrier assembly according to claim 2, wherein said lock receiving member comprises an outer curved abutment that couples to said resting surface, said outer curved abutment being curved to correspond to a curvature of said link members of said folding bicycle lock.

5. The bicycle lock carrier assembly according to claim 1, wherein said lock receiving member has an inner, hollow, circular chamber for receiving therein said lock of said bicycle lock.

6. The bicycle lock carrier assembly according to claim 1, wherein said resting surface comprises one or more lateral lock stop members.

7. The bicycle lock carrier assembly according to claim 1, wherein said resting surface comprises one or more bicycle frame stop members.

8. The bicycle lock carrier assembly according to claim 1, wherein when said bicycle lock is placed on said resting surface and locked to said lock receiving member, said bicycle lock overlies, and prevents access to, said one or more mounting provisions.

9. A bicycle lock carrier assembly comprising: a bicycle lock carrier comprising a resting surface that comprises a lock receiving member and one or more mounting provisions for securing to a frame of a bicycle; and a bicycle lock comprising a lock that fits inside said lock receiving member, wherein said bicycle lock has a configuration in which said bicycle lock is locked to said bicycle lock carrier and blocks access to said one or more mounting provisions.

10. The bicycle lock carrier assembly according to claim 9, wherein said bicycle lock comprises a lock that fits inside said lock receiving member, said lock comprising a lock member that is arranged to engage a locking structure of a portion of said bicycle lock which is outside said lock receiving member when said bicycle lock is placed on said resting surface and overlies said one or more mounting provisions, so as to lock said bicycle lock to said lock receiving member of said bicycle lock carrier.